ACS APPLIED ENERGY MATERIALS

Subscriber access provided by Kaohsiung Medical University

Article

Design of Palladium-Doped g-C3N4 for Enhanced Photocatalytic Activity towards Hydrogen Evolution Reaction

Nan Wang, Jing Wang, Jinhui Hu, Xiaoqing Lu, Jie Sun, Feng Shi, Zong-Huai Liu, Zhibin Lei, and Ruibin Jiang

ACS Appl. Energy Mater., **Just Accepted Manuscript** • DOI: 10.1021/acsaem.8b00526 • Publication Date (Web): 22 May 2018

Downloaded from http://pubs.acs.org on May 23, 2018

Just Accepted

"Just Accepted" manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides "Just Accepted" as a service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. "Just Accepted" manuscripts appear in full in PDF format accompanied by an HTML abstract. "Just Accepted" manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are citable by the Digital Object Identifier (DOI®). "Just Accepted" is an optional service offered to authors. Therefore, the "Just Accepted" Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the "Just Accepted" Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these "Just Accepted" manuscripts.

ACS Publications

is published by the American Chemical Society. 1155 Sixteenth Street N.W., Washington, DC 20036

Published by American Chemical Society. Copyright © American Chemical Society. However, no copyright claim is made to original U.S. Government works, or works produced by employees of any Commonwealth realm Crown government in the course of their duties.

Design of Palladium-Doped g-C_3N_4 for Enhanced Photocatalytic Activity towards Hydrogen Evolution Reaction

Nan Wang,[†] *Jing Wang,*[†] *Jinhui Hu,*[†] *Xiaoqing Lu,**[‡] *Jie Sun,*[†] *Feng Shi,*[†] *Zong-Huai Liu,*[†] *Zhibin Lei,*[†] *and Ruibin Jiang**[†]

[†]Shaanxi Key Laboratory for Advanced Energy Devices, Shaanxi Engineering Lab for Advanced Energy Technology, School of Materials Science and Engineering, Shaanxi Normal University, Xi'an 710119, China

[‡]College of Science, China University of Petroleum, Qingdao, Shandong 266582, China

KEYWORDS: graphitic carbon nitride, palladium doping, hydrogen evolution, photcatalysis, catalytic kinetics

ABSTRACT: Graphitic carbon nitride (g-C_3N_4) has been believed as a promising photocatalyst for water splitting owing to its right band gap and band edges. However, the kinetics of hydrogen evolution on g-C_3N_4 is very slow. Co-catalysts are usually needed to improve the catalytic kinetics. Herein, palladium-doped graphitic carbon nitride (g-C_3N_4-Pd) is designed by virtue of the tenaciously coordination of Pd atoms with the pyridinic nitrogen atoms of sixfold cavities in

g-C_3N_4. The introduction of Pd does not affect the structure and morphology of g-C_3N_4. Palladium is found to exist as Pd ions in g-C_3N_4-Pd catalysts. g-C_3N_4-Pd catalysts exhibit clearly higher hydrogen evolution activities than g-C_3N_4. The highest hydrogen evolution activity on g-C_3N_4-Pd is 15.3 times of that of g-C_3N_4. The improvement of hydrogen evolution activity is found to arise from both the alternation of the electron excitation manner and the acceleration of hydrogen evolution kinetics induced by Pd doping. Our findings provide a promising way to improve the photocatalytic performance for hydrogen evolution and pave a new avenue for the development of highly efficient and cost-effective photocatalysts for water splitting.

INTRODUCTION

Photocatalytic water splitting on semiconductors is a green and friendly route to directly convert solar energy into renewable and clean hydrogen energy.[1] Fully photocatalytic water splitting requires semiconductors with good stability and appropriate band gap and edges. The band gap must be larger than 1.23 eV but not too large, because too large band gap would limit the utilization of solar energy.[2-4] On the other hand, the conduction (valence) band edge must be more negative (positive) than the redox potential of H^+/H_2 (O_2/H_2O).[2-4] Although many semiconductors have been explored for photocatalytic water splitting,[5-11] most of them either are instable or have improper band structure.[9-13] For example, TiO_2 is stable but has very large band gap and therefore is inactive in the visible and near infrared regions.[5,6] CdS has proper band gap and conduction and valence band edges while it can be oxidized during photocatalysis.[7,8]

The g-C_3N_4, which consists of two dimensional conjugated planes packed together with tri-s-triazine repeating units through interlayer van der Waals interactions, has attracted ever-growing attention owing to its high stability, nontoxicity, and visible-light response.[14-20] The g-C_3N_4 possesses a bandgap of ~2.7 eV, and its conduction and valence band edges are ~-1.1 V

and 1.6 V versus normal hydrogen electrode, respectively,[14] which is thus qualified for water splitting in the visible region. However, the kinetics of g-C_3N_4 for photocatalytic water splitting is very slow because of the fast electron-hole recombination rate and slow hydrogen and oxygen evolution rate. The catalytic active site for hydrogen evolution on g-C_3N_4 is believed to be the pyridinic N atoms.[21,22] The relatively strong binding between the pyridinic nitrogen atoms of g-C_3N_4 and hydrogen atoms results in slow kinetics for hydrogen evolution. As a result, co-catalysts are generally needed to realize high photocatalytic performance for water splitting.[14,23-31] The co-catalysts for oxygen evolution are usually noble metal oxides, for example IrO_x and RhO_x.[14,30,31] The co-catalysts for hydrogen evolution are generally noble metal nanoparticles.[14,23-30] For example, Ag has been used as co-catalysts to enhance the photocatalytic activity of g-C_3N_4 for hydrogen evolution.[27] Besides water splitting, co-catalysts are also employed to enhanced the photocatalytic activity towards NO oxidation and CO_2 reduction.[33,34] Generally, downsizing noble metals to clusters or even single atoms provides an effective way to maximize the atom efficiency and catalytic activity.[32,35] Fortunately, the tri-s-triazine structure of g-C_3N_4 facilitates the binding or intercalation of exotic atoms in the matric of g-C_3N_4,[36-40] providing a potential scaffold for firmly trapping the highly active single metal atoms as high-efficiency photocatalytic systems for hydrogen evolution,[38] CO_2 reduction,[39] and hydrogenations.[37]

Here, by virtue of the tenaciously interaction between Pd atoms with the pyridinic nitrogen atoms of sixfold cavities in g-C_3N_4, we design Pd-doped g-C_3N_4 (g-C_3N_4-Pd) for improving the photocatalytic kinetics of hydrogen evolution reaction. On the one hand, the coordination Pd with the pyridinic nitrogen atoms may create excitation of electrons from g-C_3N_4 to Pd, which increases the separation of electron-hole pairs. On the other hand, the doped Pd can accelerate the hydrogen evolution kinetics. We find that the Pd doping does not induce obvious change of

the main structure of g-C_3N_4, while enhances the photocatalytic performance for hydrogen evolution by 15 times. Further experimental and theoretical studies confirm our speculation that the Pd doping alters the excitation manner of electrons and accelerates the kinetics of hydrogen evolution, which enhances the photocatalytic activity for hydrogen evolution.

RESULTS AND DISCUSSION

The g-C_3N_4-Pd photocatalysts were prepared through the directly pyrolysis of the mixture of dicyandiamide and H_2PdCl_4 (Figure 1a). The mixture was prepared by the dissolution of dicyandiamide and H_2PdCl_4 in deionized water. During the stirring at 80 °C, the chlorions are replaced by dicyandiamide because of the stronger interaction between Pd ions and imino group, which is confirmed by the color change of the solution (Figure S1 in the Supporting Information). The mixture solution was then dried by the evaporation of water. The obtained crystal powder was finally pyrolysed under argon protection at 550 °C. Figure 1b shows the high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image taken on the g-C_3N_4-Pd20, which was prepared by adding dicyandiamide (2 g) into H_2PdCl_4 aqueous solution (20 mL, 0.01 M). No Pd nanoparticles can be observed from low and high resolution HAADF-STEM images (Figure 1b and Figure S2 in the Supporting Information). On the high resolution HAADF-STEM image, stripes with space of 1.09 nm are clearly observed, indicating that Pd-doping makes part of g-C_3N_4 with large interlayer space. The energy-dispersive X-ray (EDX) elemental mapping reveals that the distributions of C, N and Pd are very uniform in the sample (Figure 1c–e). Since no Pd nanoparticles are observed on the HAADA-STEM image but Pd distribution is observed on the elemental mapping, the Pd may exist as single atoms. Previous studies have shown that Pd can exist stably as single atoms on g-C_3N_4 and the most stable site for Pd ions is the sixfold cavity.[39]

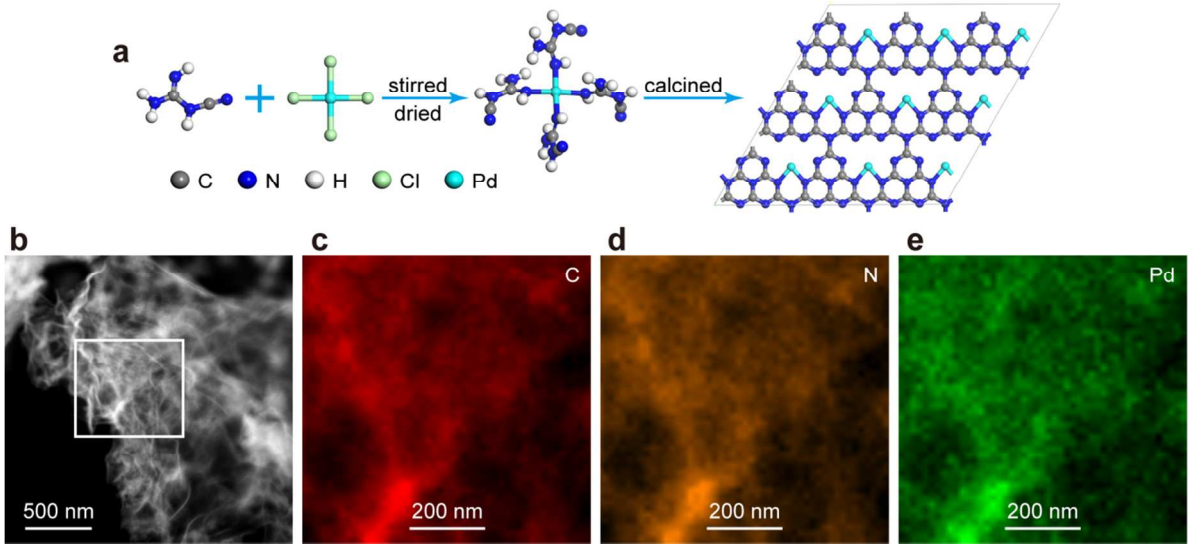

Figure 1. a) Schematic of synthesis of g-C_3N_4-Pd phototcatalysts. b) HAADF-STEM image of g-C_3N_4-Pd20. c–e) Elemental maps of C, N and Pd in the region shown in (b), respectively.

The contents of Pd can be tuned by varying the amount of H_2PdCl_4 used in the preparation of precursors. We have prepared four catalyst samples by using H_2PdCl_4 aqueous solution (0.01 M) of 0, 10, 20, and 30 mL, which are named as g-C_3N_4, g-C_3N_4-Pd10, g-C_3N_4-Pd20, and g-C_3N_4-Pd30, respectively. The Pd doping contents of g-C_3N_4-Pd10, g-C_3N_4-Pd20, and g-C_3N_4-Pd30 are 0.9, 1.2, and 1.9 wt.%, respectively, obtained from inductively coupled plasma optical emission spectrometer (ICP-OES) (Figure S3 in the Supporting Information). Their transmission electron microscopy (TEM) images are shown in Figure S4 in the Supporting Information. No Pd nanoparticles can be observed for the four samples. The HAADF-STEM image taken on the g-C_3N_4-Pd30 also indicates that no Pd nanoparticles are produced (Figure S5a in the Supporting Information). These results indicate that the Pd may exist as single atoms. Similar to g-C_3N_4-Pd20, stripes with space of 1.09 nm are also observed on STEM image of g-C_3N_4-Pd30 (Figure S5a in the Supporting Information), indicating the layer space of a part of g-C_3N_4 is increased by Pd-doping. The Brunauer-Emmett-Teller characterization indicates that the specific surface areas of the four samples are smaller than 5 m^2/g and the four samples are not porous structures

(Figure S6 in the Supporting Information), which is consistent with previous study on g-C_3N_4 prepared by direct hydrolysis of dicyandiamide.[41]

Figure 2 shows the X-ray diffraction (XRD) patterns of the four samples. The sample without Pd doping, g-C_3N_4, exhibits two diffraction peaks at around 13.5° and 27.4°, which is consistent with bulk g-C_3N_4 (JCPDS 87-1526). These two peaks are still preserved in the XRD patterns of other three samples (Figure 2a), indicating that the original crystal structure of g-C_3N_4 is largely retained after Pd doping. For g-C_3N_4-Pd20 and g-C_3N_4-Pd30, a new peak located at 8.3° is observed. The corresponding distance is 10.6 Å, which is very close to the space of stripes observed on HAADF-STEM images. Therefore, this peak arises from the diffraction of the part of g-C_3N_4 that interlayer space is increased by Pd-doping. No diffraction peaks related to Pd nanoparticles are observed on the XRD patterns, confirming that there is no Pd nanoparticles formed again. The Fourier-transform infrared spectra of the four samples are also confirmed that the Pd doping does not bring obvious change of the crystal structure of g-C_3N_4 (Figure S7 in the Supporting Information).

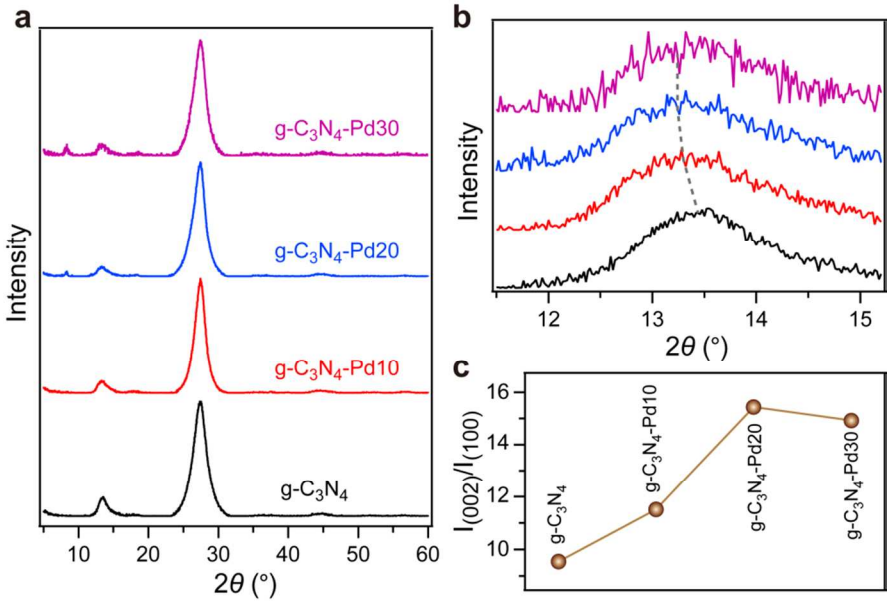

Figure 2. XRD characterization of g-C_3N_4 and Pd-doped C_3N_4 samples. a) Full XRD patterns. b) Enlarged view of (100) peaks. The colors of the lines in (b) have the same designations as in (a). c) Intensity ratios of (002) to (100) peaks of the four samples.

Although no obvious change has occurred to the matrix network of Pd-doped g-C_3N_4, the subtle alteration in the crystal structure can be unveiled by the detailed analysis on the two diffraction peaks. The peak at around 13.5° corresponds to the diffraction from (100) facet of g-C_3N_4, which reflects the in-plane structural packing motif of the heptazine units.[20,42] As the doping content of Pd is increased, the (100) peak first shifts toward low angle and then keeps at a fixed angle (Figure 2b). The shift of (100) peak toward low angle indicates that the Pd doping leads to the increase of the distance of in-planar nitride pores. In contrast, the Pd doping does not induce observable shift of the peak at around 27.4° (Figure S8 in the Supporting Information), which corresponds to the diffraction from (002) facets and reflects the interlayer distance of g-C_3N_4. No observable shift of (002) peak indicates that the Pd doping does not change interlayer distance. Previous studies have indicated that interlayer distance is increased when the doping atoms or groups locate at the surface of g-C_3N_4 instead of sixfold cavities.[19,24] In our Pd-doped g-C_3N_4, the distance of in-planar nitride pores is increased while the interlayer distance is not altered, definitely indicating that the Pd atoms locate at the sixfold cavities of g-C_3N_4. The intensity ratio between (100) and (002) peaks were also investigated (Figure 2c). It first increases with the increase of Pd doping content and then decreases. This variation trend can be understood by the changes of in-plane periodicity of g-C_3N_4 induced by Pd doping. When the doping content of Pd is small, Pd atoms are randomly distributed into the sixfold cavities of g-C_3N_4, breaking the in-plane periodicity. The diffraction intensity of (100) facet is therefore reduced. As Pd doping content is further increased, most of sixfold cavities of g-C_3N_4 are occupied by Pd atoms. The in-plane periodicity is gradually recovered and thereby the diffraction

intensity of (100) is increased. For comparison, we have also prepared g-C_3N_4 loaded with Pd nanoparticles (Figure S9 in the Supporting Information). Different from Pd-doped g-C_3N_4, the Pd-nanoparticle-loaded g-C_3N_4 exhibits a very similar XRD pattern to that of the bare g-C_3N_4 (Figure S10 in the Supporting Information). Two diffraction peaks does not show observable shifts. The relative intensity of (100) diffraction peak is not reduced after Pd nanoparticle loading.

X-ray photoelectron spectroscopy (XPS) was carried out on g-C_3N_4 and g-C_3N_4-Pd20 to reveal the chemical composition and states. On the wide XPS spectra (Figure 3a), only peaks corresponding to the binding energy of 1s electrons of C, N, O are observed on g-C_3N_4. The O should come from the surface adsorbed adventitious oxygen-containing species. Besides the C, N and O 1s peaks, the peaks corresponding to Pd 3d are also observed on the XPS spectrum of g-C_3N_4-Pd20, indicating that no other elements are contained in g-C_3N_4-Pd20. The molar ratios of C and N in both samples are close to 3:4, which is consistent with the stoichiometric composition of g-C_3N_4. Further analysis on the C 1s peak shows that in the two samples C 1s contains two peaks, located at 288.2 and 284.8 eV (Figure 3b). The peak at 288.2 eV corresponds to the C atoms of N–C–N_2.[20,43] The peak at 284.8 eV is related to the C atoms bound with C or H atoms, which may arise from the edges or defects.[20,43] The XPS spectra of N 1s are almost identical for g-C_3N_4 and g-C_3N_4-Pd20 (Figure 3c). They can be decomposed into four peaks, which are located at 398.7, 400.1, 401.4 and 404.3 eV. They can be ascribed to N atoms of C–N–C, tertiary nitrogen N–$(C)_3$, N–H groups, and π-excitations from low to high energy, respectively.[20,43] The XPS spectra of both C and N atoms clearly indicate that the Pd doping brings negligible changes on the chemical states of C and N atoms. This is consistent with previous theoretical results that the electron transfer between Pd ions and the g-C_3N_4 is very weak when Pd ions locate at the sixfold cavities.[39] For XPS spectrum of Pd in g-C_3N_4-Pd20, the $3d_{3/2}$ and $3d_{5/2}$ are located at

343.6 and 338.3 eV respectively (Figure 3d). In contrast, the $3d_{3/2}$ and $3d_{5/2}$ in Pd nanoparticles are located at 340.7 and 335.5 eV, respectively. The clearly shift toward higher binding energy of Pd 3d indicates that Pd exists as Pd^{2+} ions in g-C_3N_4. Similar result is also obtained on Pd 3d in g-C_3N_4-Pd30 sample (Figure S5b in the Supporting Information). Moreover, the XPS spectrum of O 1s in g-C_3N_4-Pd20 is nearly identical to that in g-C_3N_4 (Figure S11 in the Supporting Information). Especially, no peak appears at ~528−531 eV in O 1s XPS spectrum in g-C_3N_4-Pd20, which corresponds to O in metal oxides.[44] Therefore, the Pd^{2+} ions do not originate from the palladium oxide.

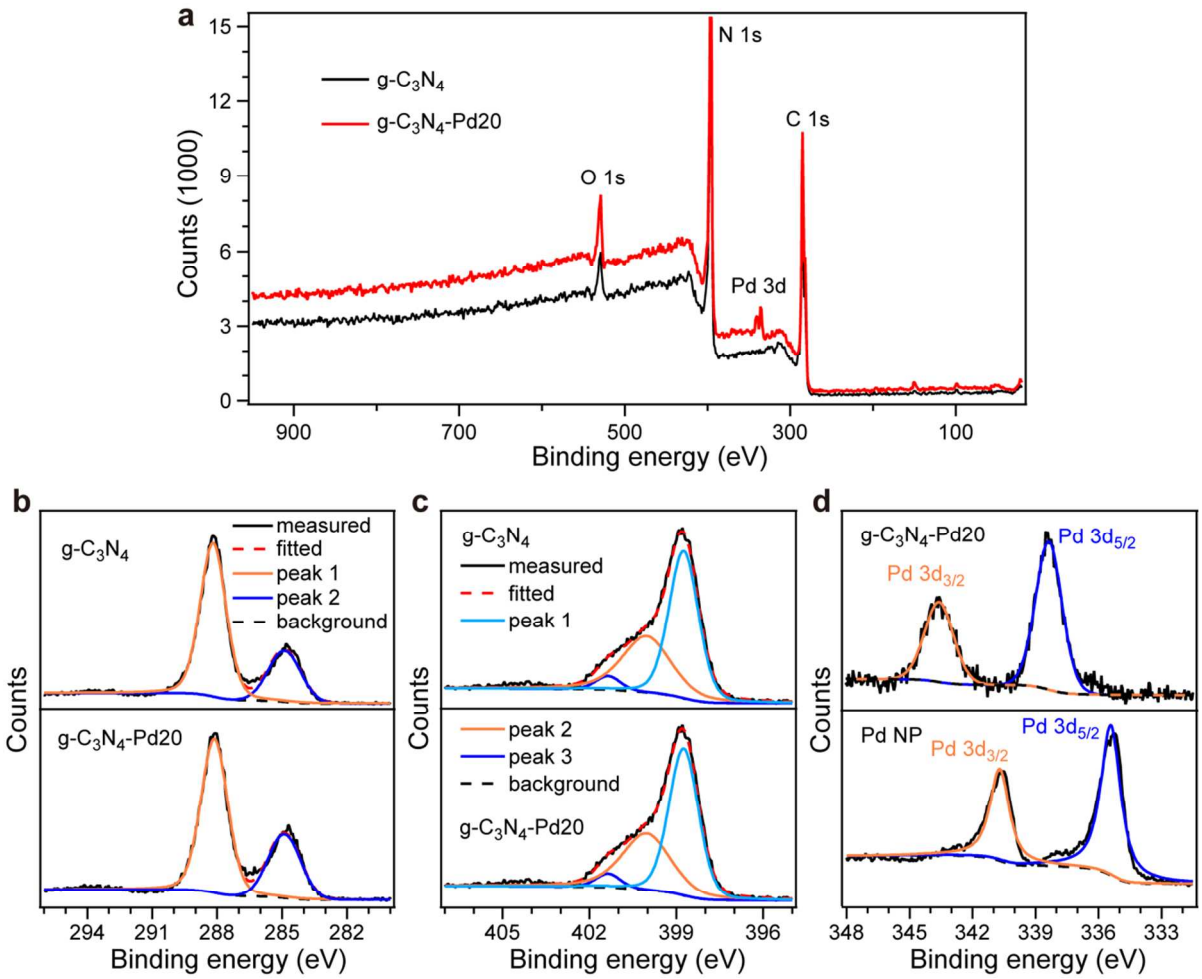

Figure 3. XPS characterization of g-C_3N_4 and g-C_3N_4-Pd20 samples. a) Wide XPS spectra. b) C 1s spectra. c) N 1s spectra. d) Pd 3d spectra.

Figure 4a shows the absorbance of the g-C_3N_4 samples with different Pd doping contents. Clearly, the Pd doping extends the absorption of g-C_3N_4 toward longer wavelength. The increase of absorption at longer wavelength is directly reflected by color of the samples. As the Pd doping content is increased, the color of the samples changes from yellow to light brown and to dark brown. The absorption at longer wavelength indicates that the Pd doping forms midgap energy levels. Under light excitation, electrons transfer from g-C_3N_4 to the doped Pd ions, which is similar to the ligand-to-metal transfer in complex compound and confirmed by the following density functional theory (DFT) calculations. The extended and enhanced absorption of g-C_3N_4 in the longer wavelength is highly desired for photocatalytic water splitting.

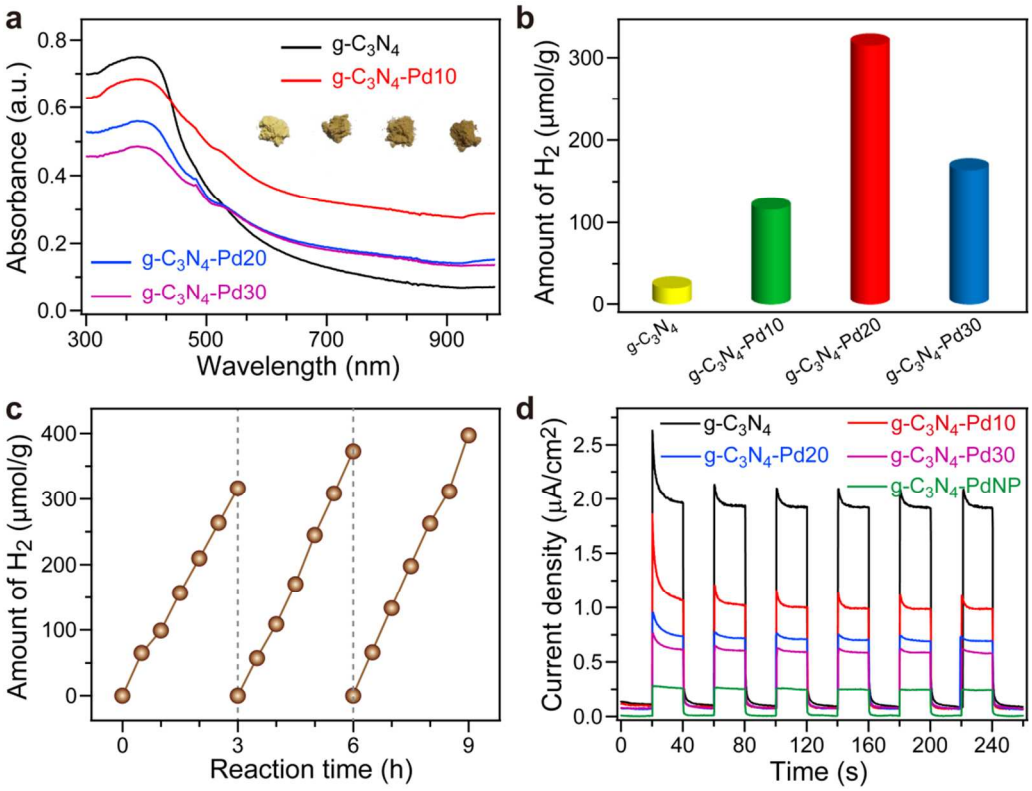

Figure 4. a) Absorption spectra of g-C_3N_4 and differently Pd-doped g-C_3N_4 samples. Inset is the photographs of the g-C_3N_4, g-C_3N_4-Pd10, g-C_3N_4-Pd20, and g-C_3N_4-Pd30 samples from left to right. b) Amount of hydrogen produced within three hours for the four samples. c) Cycling photocatalytic H_2 production activity of g-C_3N_4-Pd20. d) Photoelectrochemical current of g-C_3N_4, g-C_3N_4-Pd10, g-C_3N_4-Pd20, g-C_3N_4-Pd30, and g-C_3N_4-PdNP samples.

The photocatalytic hydrogen evolution performance of the Pd-doped g-C_3N_4 is evaluated under visible light (λ > 420 nm) illumination. Figure 4b gives the amount of hydrogen produced by the g-C_3N_4 doped with different Pd contents within three hours. The Pd-doping clearly improved the photocatalytic hydrogen evolution performance of g-C_3N_4. The performance for hydrogen evolution first increases with the increase of Pd doping amount and then decreases. The g-C_3N_4-Pd20 gives the highest photocatalytic activity for hydrogen evolution. The amount of hydrogen produced by g-C_3N_4-Pd20 within three hours is 316.2 μmol/g, which is 15.3 times of the amount of hydrogen produced by g-C_3N_4 without Pd doping. The quantum efficiencies of the catalysts were also investigated at 405 nm. At the irradiation of 405-nm light-emitting diode, no hydrogen is detected on g-C_3N_4 within 4 h. Therefore, the quantum efficiency of g-C_3N_4 is not obtained. The quantum efficiencies of g-C_3N_4-Pd10, g-C_3N_4-Pd20, and g-C_3N_4-Pd30 are 0.22%, 0.28%, 0.13% at 405 nm, respectively (Figure S12 in the Supporting Information). To study whether the Pd doping affect the stability of g-C_3N_4 during photocatalysis, we carried out cyclical stability test on g-C_3N_4-Pd20. During three cycles, the catalyst does not show any degradation in photocatalytic performance (Figure 4c), indicating that the Pd doped g-C_3N_4 has very good stability during photocatalytic hydrogen evolution. In contrast, the photocatalytic activity increases in the three cycles. The increase of photocatalytic activity can be attributed to the fact that the dispersion of the catalyst is improved as the photocatalytic reaction takes place. In addition, we have also prepared g-C_3N_4-Pt, g-C_3N_4-Au, and g-C_3N_4-Ag catalysts by changing H_2PdCl_4 to H_2PtCl_6, $HAuCl_4$, and $AgNO_3$. No Pt and Ag nanoparticles are observed on the g-C_3N_4, while Au nanoparticles are observed (Figure S13 in the Supporting Information). The photocatalytic activities of g-C_3N_4-Pt, g-C_3N_4-Au, and g-C_3N_4-Ag towards hydrogen evolution are lower than that of g-C_3N_4-Pd (Figure S14 in the Supporting Information).

The photoelectrochemical performance of the Pd-doped g-C_3N_4 was also investigated (Figure 4d). When light is turned on, the photoelectrochemical currents of all samples have very sharp edges and then gradually decrease to steady values. This phenomenon may arise from the fast photo-oxidation of defects or edges of g-C_3N_4. Unlike hydrogen evolution, the photocurrent is reduced with the increase of Pd doping amount. The different performance of Pd-doped g-C_3N_4 in hydrogen evolution and photocurrent can be understood by the reaction difference between hydrogen evolution and photocurrent. In hydrogen evolution reaction, the reduction and oxidation reactions simultaneously occur on the Pd-doped g-C_3N_4 (Figure S15a in the Supporting Information). The Pd doping accelerates the electron-hole separation and the kinetics of hydrogen evolution, leading to the improvement of the photocatalytic performance. In contrast, in the photocurrent measurements, g-C_3N_4 acts as photoanode because it is a n-type semiconductor. Only oxidation reaction takes place on the electrode. The advantage of Pd-doped g-C_3N_4, improving kinetics of hydrogen evolution, cannot be reflected. Contrarily, Pd ions act as trap centers for electrons in the photoelectrode and therefore inhibit the transfer of electrons to F-doped SnO_2-coated (FTO) glass and increase the electron-hole recombination rate (Figure S15b in the Supporting Information). As a result, the Pd doping reduces the photoelectrochemical current of g-C_3N_4. Because of the same reason, the g-C_3N_4-PdNP also exhibits smaller photoelectrochemical current than g-C_3N_4.

In order to confirm our deduction, we carried out DFT calculations on g-C_3N_4 with different Pd doping contents. The doping sites of Pd were first studied. It is found that only the Pd doped at the cavity site is stable. When Pd is set at other sites, Pd will move to the cavity site after the geometry optimization. Therefore, the Pd doped at the cavity site is employed in the following DFT calculations. Figure 5 displays the highest occupied orbitals (HOMO) and the lowest

unoccupied orbitals (LUMO) of the g-C_3N_4 and Pd-doped g-C_3N_4. In the calculations, we considered 25%, 50%, 75% and 100% sixfold cavities of g-C_3N_4 are occupied by Pd atoms. For simplicity, they are named as Pd-25%, Pd-50%, Pd-75% and Pd-100%, respectively. The HOMO and LUMO orbitals are near uniformly distributed on g-C_3N_4 (Figure 5a). The HOMO orbital of Pd-25% mainly locates at g-C_3N_4, while the LUMO is mainly distributed on Pd atom (Figure 5b). Such an HOMO and LUMO distributions mean that electrons are transferred from g-C_3N_4 to Pd atom when excited by light, which is similar to ligand-to-metal electron transfer in complex compound. Such an electron transfer under excitation facilitates the electron-hole separation, which is in favor of photocatalysis. Similar to Pd-25%, the HOMO and LUMO orbitals of Pd-50% are mainly distributed at g-C_3N_4 and Pd atoms, respectively (Figure 5c). Therefore, Pd-50% also favors the photocatalysis. In contrast, both HOMO and LUMO of Pd-75% and Pd-100% mainly locate at the doped Pd atoms (Figure 5d and e), which is not in favor of electron-hole separation and thereby not helpful to the photocatalysis. The variation of HOMO and LUMO with the Pd doping contents explains the change of hydrogen evolution activity of g-C_3N_4 with the Pd doping contents.

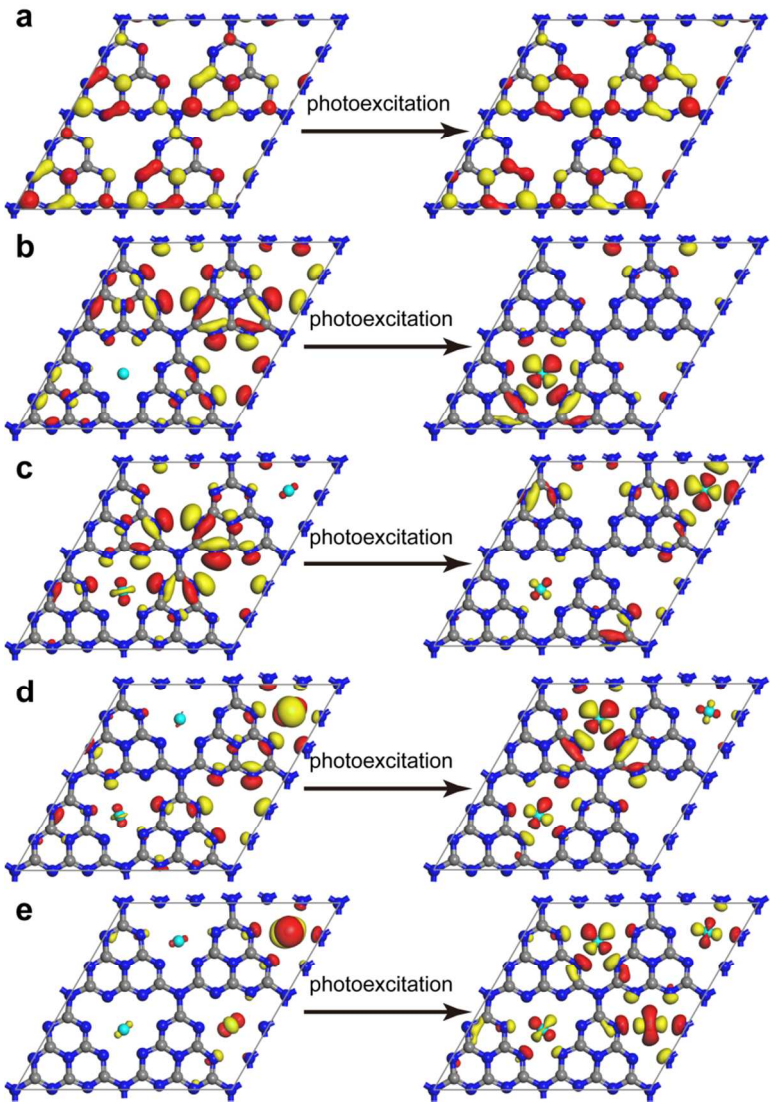

Figure 5. HOMO and LUMO of g-C_3N_4 and differently Pd-doped g-C_3N_4. a) g-C_3N_4. b–e) 25%, 50%, 75% and 100% sixfold cavities of g-C_3N_4 occupied by Pd atoms. The yellow and red bubbles represent the positive and negative of orbitals. The isovalue is set at ± 0.03 e $Å^{-3}$.

The hydrogen atom adsorption on g-C_3N_4 and Pd-doped g-C_3N_4 was further studied to unravel the effect of Pd doping on the hydrogen evolution kinetics. On g-C_3N_4, hydrogen atom can adsorb stably at the nitrogen atoms of the sixfold cavities with very large adsorption energy of 4.00 eV (Figure 6a). Hydrogen atoms can adsorb stably on Pd atoms of Pd-doped g-C_3N_4 (Figure 6b). Compared with g-C_3N_4, the adsorption energy of hydrogen on Pd atoms is greatly reduced to 2.11 eV. The hydrogen adsorption Gibbs free-energy (ΔG_H) is a widely accepted

descriptor of hydrogen evolution activity for various catalytic materials, where the optimal value of ΔG_H is around zero eV to compromise the reaction barriers and achieve the best hydrogen evolution activity.[45-47] We therefore calculated the Gibbs free-energy of the hydrogen adsorbed on g-C_3N_4 and Pd-doped g-C_3N_4. As shown in Figure 6c, the Gibbs free-energy of hydrogen adsorbed on Pd is greatly reduced to -1.82 eV compared with hydrogen adsorbed on g-C_3N_4 (-3.71 eV). The greatly reduction of Gibbs free-energy indicates that the Pd doping improves the hydrogen evolution kinetics on g-C_3N_4. On the basis of above studies, we believe that the improvement of hydrogen evolution performance by Pd doping arises from the alternation of electron excitation manner and the improvement of hydrogen evolution kinetics. The Pd doping results in electrons excited directly from g-C_3N_4 to the doped Pd atoms which are just the active sites for hydrogen evolution (Figure 6d). This mechanism is different from the improvement mechanism of metal nanoparticles loaded on g-C_3N_4. In g-C_3N_4 loaded with metal nanoparticles, electrons are first excited into conduction band of g-C_3N_4 and then transfer to metal nanoparticles. The transfer of excited electrons to metal nanoparticles needs to diffuse a certain distance, which would increase the electron-hole recombination.

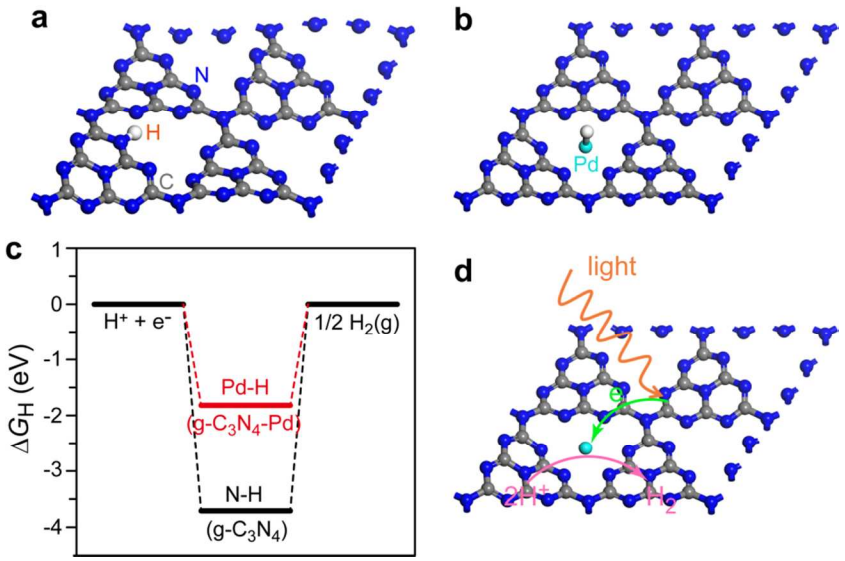

Figure 6. a) Adsorption configuration of H on g-C_3N_4. b) Adsorption configuration of H on Pd-doped g-C_3N_4. c) Free-energy diagram of H adsorbed on g-C_3N_4 and differently Pd-doped g-C_3N_4 samples. d) Mechanism of photocatalytic hydrogen evolution in Pd-doped g-C_3N_4.

CONCLUSION

In conclusion, we have prepared Pd-doped g-C_3N_4 and studied their photocatalytic hydrogen evolution performances. The Pd doping extends the absorption of g-C_3N_4 toward longer wavelength through forming excitation from g-C_3N_4 to Pd atoms. The performance of photocatalytic hydrogen evolution of g-C_3N_4 is improved by 15.3 times by the Pd doping. The improvement of photocatalytic hydrogen evolution is found to arise from the alternation of electron excitation manner and the improvement of hydrogen evolution kinetics. Our findings provide a promising way to improve the photocatalytic performance for hydrogen evolution and pave a new avenue for the development of highly efficient and cost-effective photocatalysts for water splitting.

EXPERIMENTAL SECTION

All the chemicals were of analytical grade and used as-received without further purification. Deionized water was used throughout the experiment.

Preparation of catalysts. The synthesis of g-C_3N_4-Pd was carried out by modifying the previously reported method for g-C_3N_4.[48] The samples were prepared by thermal condensation. Dicyandiamide (2 g) was added into different volumes of H_2PdCl_4 aqueous solution (0.01 M). After stirred at 80 °C for 3 hours, the solution was dried at 80 °C. The obtained powder was ground in an agate mortar. The ground powder was then transferred into an alumina boat crucible and heated to 550 °C at a heating rate of 5 °C min^{-1} and maintained for 4 h under flowing argon atmosphere. The g-C_3N_4-Pd10, g-C_3N_4-Pd20 and g-C_3N_4-Pd30 were prepared by adding

dicyandiamide into 10, 20 and 30 mL H_2PdCl_4 solution. After the reaction, the product was ground to obtain a fine g-C_3N_4-Pd powder. For comparison, g-C_3N_4 was prepared following the same protocol mentioned above except that the H_2PdCl_4 was replaced by 20 mL of water. In addition, the g-C_3N_4-Pt, g-C_3N_4-Au and g-C_3N_4-Ag were prepared by using the similar method. The only difference was that the H_2PdCl_4 was changed into H_2PtCl_6, $HAuCl_4$, and $AgNO_3$.

Characterization. TEM imaging was carried out on a JEM-2100 microscope (JEOL) operating at 200 kV. HAADF-STEM imaging and energy-dispersive X-ray elemental mapping were obtained on an FEI Tecnai F20 microscope equipped with an Oxford energy-dispersive X-ray analysis system. XRD patterns were conducted on a Rigaku Smartlab diffractometer with Cu Kα (λ = 1.5406 Å) radiation. UV-Vis diffuse reflectance spectra were recorded for the dry pressed disk samples by using a UV-Vis spectrophotometer (UV-Lambda 950, PerkinElmer, America), and $BaSO_4$ was used as a reference standard. FTIR spectra were performed in transmission mode from 4000 to 400 cm^{-1} on a Bruker Tensor 27 instrument using the KBr pellet technique. XPS data were collected on an Axis Ultra spectrometer (Kratos Analytical) equipped with a monochromatized Al Kα X-ray source (1486.6 eV). The binding energy was corrected by the C 1s line at 284.6 eV. The Pd doping contents were measured on an Inductively Coupled Plasma (ICP-OES) instrument (Prodigy 7, Teledyne Leeman Labs, America). Microwave digestion technique was employed to dissolve samples. A certain amount of catalyst was first dissolved in a mixed solution containing concentrated sulfuric acid and nitric acid (1:4 in volume). The mixture was then treated by 400-W microwave (MDS-8G, Sineo, China) at 120 °C for 10 min. Surface areas and pore size distributions were measured by nitrogen adsorption and desorption at 77 K on an ASAP 2020HD88 system (Micromeritics Instrument Corporation).

Photocatalytic Hydrogen Evolution Test. The photocatalytic hydrogen evolution was carried out in a closed circulation system using a Pyrex cell (total volume about 250 mL with a top irradiation area of around 25 cm^2). The photocatalysts (100 mg) were dispersed in triethanolamine aqueous solution (10 vol%, 100 mL). A 300 W xenon lamp (PLS-SXE 300, PerfectLight) equipped with a 420 nm long-pass filter was used as the visible light source. The reaction cell was kept at 7 °C with cooling water. The amount of evolved hydrogen was determined using an online gas chromatograph (GC-7806, Shiweipx, TCD detector, nitrogen as the carrier gas, and 5 Å molecular sieve column).

Photoelectrochemical Measurement. Time-dependent photocurrents were measured on an electrochemical analyzer (CHI 660D Instruments) with a standard three-electrode system. The photocatalysts coated on FTO glass was used as the working electrodes. A Pt wire was employed as the counter electrode. The reference electrode was Ag/AgCl (saturated KCl). The working electrodes were prepared by dispersing photocatalysts (1.8 mg) into 1 mL of N,N-dimethylformamide to get a slurry and then spreading the slurry onto FTO glass. The slurry cover FTO glass was dried at 80 °C, followed by heated at 300 °C for 2 h in a muffle furnace to improve adhesion. The electrolyte was Na_2SO_4 (0.5 M). The light source was a 300 W xenon lamp (PLS-SXE 300, PerfectLight) equipped with a 420 nm long-pass filter.

DFT Calculations. DFT calculations were performed with the program package DMol in Materials Studio of Accelrys Inc. The exchange correlation energy was treated by the Perdew−Burke−Ernzerhof (PBE) functional based on the generalized gradient approximation (GGA).[49] In order to consider dispersion force, a semiempirical DFT-D2 method proposed by Grimme was exploited.[50] The valence electron functions were expanded into a set of numerical atomic orbitals by a double numerical basis with polarization functions (DNP). A g-C_3N_4 plane containing 2 × 2

unit cell was used as the model. A vacuum of 20 Å was used to eliminate interlayer interactions. The 6 × 6 × 1 *k*-points was performed for the first Brillouin zone using the gamma center scheme. The hydrogen adsorption free energies, ΔG_{H}, were determined according to equations 1 and 2:

$$\Delta E_{\mathrm{H}} = E_{(\mathrm{catalyst+H})} - E_{(\mathrm{catalyst})} - \frac{1}{2} E_{(\mathrm{H_2})} \quad (1)$$

where $E_{(\mathrm{catalyst+H})}$ and $E_{(\mathrm{catalyst})}$ refer to the energies of the catalyst with hydrogen adsorption and clean catalyst, respectively, and $E_{(\mathrm{H_2})}$ refers the energy of gas phase hydrogen molecule. The hydrogen adsorption Gibbs free-energy was calculated at zero potential and pH = 0 as:

$$\Delta G_{\mathrm{H}} = \Delta E_{\mathrm{H}} + \Delta E_{\mathrm{ZPE}} - T\Delta S \quad (2)$$

where ΔE_{H} is the hydrogen adsorption energy, ΔE_{ZPE} is the difference in zero-point energy between the adsorbed state and the gas phase, T is the temperature (300 K), and ΔS is the difference in entropy between the adsorbed state and the gas phase.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: xxx.

Photographs of solution; Additional HAADF-STEM images; ICP-OES measurements; Additional TEM images; N_2 adsorption-desorption isotherm; FTIR spectra; XRD patterns; XPS spectra; Quantum efficiency; Hydrogen evolution on g-C_3N_4-Pt, g-C_3N_4-Au, and g-C_3N_4-Ag; Schematics of photocatalytic water splitting and photoelectrochemical measurements (PDF)

AUTHOR INFORMATION

Corresponding Author

*E-mail: luxq@upc.edu.cn.

*E-mail: rbjiang@snnu.edu.cn.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (61505102 and 61775129) and Fundamental Research Funds for the Central Universities (GK201602004 and GK201703028).

REFERENCES

(1) Fujishima, A.; Honda, K. Electrochemical Photolysis of Water at a Semiconductor Electrode. *Nature* **1972**, *238*, 37-38.

(2) Chen, X.; Shen, S.; Guo, L.; Mao, S. S. Semiconductor-Based Photocatalytic Hydrogen Generation. *Chem. Rev.* **2010**, *110*, 6503-6570.

(3) Walter, M. G.; Warren, E. L.; McKone, J. R.; Boettcher, S. W.; Mi, Q.; Santori, E. A.; Lewis, N. S. Solar Water Splitting Cells. *Chem. Rev.* **2010**, *110*, 6446-6473.

(4) Tong, H.; Ouyang, S.; Bi, Y.; Umezawa, N.; Oshikiri, M.; Ye, J. Nano-Photocatalytic Materials: Possibilities and Challenges. *Adv. Mater.* **2012**, *24*, 229-251.

(5) Schneider, J.; Matsuoka, M.; Takeuchi, M.; Zhang, J.; Horiuchi, Y.; Anpo, M.; Bahnemann, D. W. Understanding TiO_2 Photocatalysis: Mechanisms and Materials. *Chem. Rev.* **2014**, *114*, 9919-9986.

(6) Nolan, M.; Iwaszuk, A.; Lucid, A. K.; Carey, J. J.; Fronzi, M. Design of Novel Visible Light Active Photocatalyst Materials: Surface Modified TiO_2. *Adv. Mater.* **2016**, *28*, 5425-5446.

(7) Shang, L.; Tong, B.; Yu, H.; Waterhouse, G. I.; Zhou, C.; Zhao, Y.; Tahir, M.; Wu, L.-Z.; Tung, C.-H.; Zhang, T. CdS Nanoparticle-Decorated Cd Nanosheets for Efficient Visible Light-Driven Photocatalytic Hydrogen Evolution. *Adv. Energy Mater.* **2016**, *6*, 1501241.

(8) Li, Q.; Li, X.; Wageh, S.; Al-Ghamdi, A.; Yu, J. CdS/Graphene Nanocomposite Photocatalysts. *Adv. Energy Mater.* **2015**, *5*, 1500010.

(9) Tran, P. D.; Wong, L. H.; Barber, J.; Loo, J. S. Recent Advances in Hybrid Photocatalysts for Solar Fuel Production. *Energy Environ. Sci.* **2012**, *5*, 5902-5918.

(10) Chen, S.; Wang, L.-W. Thermodynamic Oxidation and Reduction Potentials of Photocatalytic Semiconductors in Aqueous Solution. *Chem. Mater.* **2012**, *24*, 3659-3666.

(11) Sivula, K.; van de Krol, R. Semiconducting Materials for Photoelectrochemical Energy Conversion. *Nat. Rev. Mater.* **2016**, *1*, 15010.

(12) Guo, S.; Li, X.; Zhu, J.; Tong, T.; Wei, B. Au NPs@MoS_2 Sub-Micrometer Sphere-ZnO Nanorod Hybrid Structures for Efficient Photocatalytic Hydrogen Evolution with Excellent Stability. *Small* **2016**, *12*, 5692-5701.

(13) Li, X.; Guo, S.; Kan, C.; Zhu, J.; Tong, T.; Ke, S.; Choy, W. C.; Wei, B. Au Multimer@MoS_2 Hybrid Structures for Efficient Photocatalytical Hydrogen Production via Strongly Plasmonic Coupling Effect. *Nano Energy* **2016**, *30*, 549-558.

(14) Wang, X.; Maeda, K.; Thomas, A.; Takanabe, K.; Xin, G.; Carlsson, J. M.; Domen, K.; Antonietti, M. A Metal-Free Polymeric Photocatalyst for Hydrogen Production from Water under Visible Light. *Nat. Mater.* **2009**, *8*, 76-80.

(15) Wang, Y.; Wang, X.; Antonietti, M. Polymeric Graphitic Carbon Nitride as a Heterogeneous Organocatalyst: From Photochemistry to Multipurpose Catalysis to Sustainable Chemistry. *Angew. Chem. Int. Ed.* **2012**, *51*, 68-89.

(16) Liu, J.; Liu, Y.; Liu, N.; Han, Y.; Zhang, X.; Huang, H.; Lifshitz, Y.; Lee, S.-T.; Zhong, J.; Kang, Z. Metal-Free Efficient Photocatalyst for Stable Visible Water Splitting Via a Two-Electron Pathway. *Science* **2015**, *347*, 970-974.

(17) Zhang, G.; Lan, Z.-A.; Lin, L.; Lin, S.; Wang, X. Overall Water Splitting by Pt/g-C_3N_4 Photocatalysts without Using Sacrificial Agents. *Chem. Sci*. **2016**, *7*, 3062-3066.

(18) Fu, J.; Yu, J.; Jiang, C.; Cheng, B. g-C_3N_4-Based Heterostructured Photocatalysts. *Adv. Energy Mater.* **2017**, *7*, 1701503.

(19) Li, Y.; Ouyang, S.; Xu, H.; Wang, X.; Bi, Y.; Zhang, Y.; Ye, J. Constructing Solid-Gas-Interfacial Fenton Reaction over Alkalinized-C_3N_4 Photocatalyst to Achieve Apparent Quantum Yield of 49% at 420 nm. *J. Am. Chem. Soc.* **2016**, *138*, 13289-13297.

(20) Cao, S.; Low, J.; Yu, J.; Jaroniec, M. Polymeric Photocatalysts Based on Graphitic Carbon Nitride. *Adv. Mater.* **2015**, *27*, 2150-2176.

(21) Zheng, Y.; Jiao, Y.; Zhu, Y.; Li, L. H.; Han, Y.; Chen, Y.; Du, A.; Jaroniec, M.; Qiao, S. Z. Hydrogen Evolution by a Metal-Free Electrocatalyst. *Nat. Commun.* **2014**, *5*, 3783-3790.

(22) Gao, G.; Jiao, Y.; Ma, F.; Jiao, Y.; Waclawik, E.; Du, A. Metal-Free Graphitic Carbon Nitride as Mechano-Catalyst for Hydrogen Evolution Reaction. *J. Catal.* **2015**, *332*, 149-155.

(23) Xu, J.; Zhang, L.; Shi, R.; Zhu, Y. Chemical Exfoliation of Graphitic Carbon Nitride for Efficient Heterogeneous Photocatalysis. *J. Mater. Chem. A* **2013**, *1*, 14766-14772.

(24) Ran, J.; Ma, T. Y.; Gao, G.; Du, X.-W.; Qiao, S. Z. Porous P-Doped Graphitic Carbon Nitride Nanosheets for Synergistically Enhanced Visible-Light Photocatalytic H_2 Production. *Energy Environ. Sci.* **2015**, *8*, 3708-3717.

(25) Zhang, G.; Zhang, M.; Ye, X.; Qiu, X.; Lin, S.; Wang, X. Iodine Modified Carbon Nitride Semiconductors as Visible Light Photocatalysts for Hydrogen Evolution. *Adv. Mater.* **2014**, *26*, 805-809.

(26) Zhang, J.; Zhang, M.; Lin, L.; Wang, X. Sol Processing of Conjugated Carbon Nitride Powders for Thin-Film Fabrication. *Angew. Chem. Int. Ed.* **2015**, *54*, 6297-6301.

(27) Chen, F.; Yang, H.; Luo, W.; Wang, P.; Yu, H. Selective Adsorption of Thiocyanate Anions on Ag-Modified g-C_3N_4 for Enhanced Photocatalytic Hydrogen Evolution. *Chinese J. Catal.* **2017**, *38*, 1900-1998.

(28) Maeda, K.; Wang, X.; Nishihara, Y.; Lu, D.; Antonietti, M.; Domen, K. Photocatalytic Activities of Graphitic Carbon Nitride Powder for Water Reduction and Oxidation under Visible Light. *J. Phys. Chem. C* **2009**, *113*, 4940-4947.

(29) Yang, J.; Wang, D.; Han, H.; Li, C. Roles of Cocatalysts in Photocatalysis and Photoelectrocatalysis. *Acc. Chem. Res*. **2013**, *46*, 1900-1909.

(30) Zhang, G.; Zang, S.; Wang, X. Layered $Co(OH)_2$ Deposited Polymeric Carbon Nitrides for Photocatalytic Water Oxidation. *ACS Catal*. **2015**, *5*, 941-947.

(31) Feng, J.; Lv, F.; Zhang, W.; Li, P.; Wang, K.; Yang, C.; Wang, B.; Yang, Y.; Zhou, J.; Lin, F.; Wang, G.; Guo, S. Iridium-Based Multimetallic Porous Hollow Nanocrystals for Efficient Overall-Water-Splitting Catalysis. *Adv. Mater.* **2017**, *29*, 1703798.

(32) Li, X.; Bi, W.; Zhang, L.; Tao, S.; Chu, W.; Zhang, Q.; Luo, Y.; Wu, C.; Xie, Y. Single-Atom Pt as Co-Catalyst for Enhanced Photocatalytic H_2 Evolution. *Adv. Mater.* **2016**, *28*, 2427-2431.

(33) Ni, Z.; Dong, F.; Huang, H.; Zhang, Y. New Insights into How Pd Nanoparticles Influence the Photocatalytic Oxidation and Reduction Ability of g-C_3N_4 Nanosheets. *Catal. Sci. Technol.* **2016**, *6*, 6448-6458.

(34) Cao, S.; Li, Y.; Zhu, B.; Jaroniec, M.; Yu, J. Facet Effect of Pd Cocatalyst on Photocatalytic CO_2 Reduction over g-C_3N_4. *J. Catal.* **2017**, *349*, 208-217.

(35) Tong, T.; Zhu, B.; Jiang, C.; Cheng, B.; Yu, J. Mechanistic Insight into the Enhanced Photocatalytic Activity of Single-Atom Pt, Pd or Au-Embedded g-C_3N_4. *Appl. Surf. Sci.* **2018**, *433*, 1175-1183.

(36) Ding, Z.; Chen, X.; Antonietti, M.; Wang, X. Synthesis of Transition Metal-Modified Carbon Nitride Polymers for Selective Hydrocarbon Oxidation. *ChemSusChem* **2011**, *4*, 274-281.

(37) Vilé, G.; Albani, D.; Nachtegaal, M.; Chen, Z.; Dontsova, D.; Antonietti, M.; López, N.; Pérez-Ramírez, J. A Stable Single-Site Palladium Catalyst for Hydrogenations. *Angew. Chem. Int. Ed.* **2015**, *54*, 11265-11269.

(38) Li, Y.; Wang, Z.; Xia, T.; Ju, H.; Zhang, K.; Long, R.; Xu, Q.; Wang, C.; Song, L.; Zhu, J.; Jiang, J.; Xiong, Y. Implementing Metal-to-Ligand Charge Transfer in Organic Semiconductor for Improved Visible-Near-Infrared Photocatalysis. *Adv. Mater.* **2016**, *28*, 6959-6965.

(39) Gao, G.; Jiao, Y.; Waclawik, E. R.; Du, A. Single Atom (Pd/Pt) Supported on Graphitic Carbon Nitride as an Efficient Photocatalyst for Visible-Light Reduction of Carbon Dioxide. *J. Am. Chem. Soc.* **2016**, *138*, 6292-6297.

(40) Zheng, Y.; Jiao, Y.; Zhu, Y.; Cai, Q.; Vasileff, A.; Li, L. H.; Han, Y.; Chen, Y.; Qiao, S.-Z. Molecule-Level g-C_3N_4 Coordinated Transition Metals as a New Class of Electrocatalysts for Oxygen Electrode Reactions. *J. Am. Chem. Soc.* **2017**, *139*, 3336-3339.

(41) Zhang, Y.; Liu, J.; Wu, G.; Chen, W. Porous Graphitic Carbon Nitride Synthesized via Direct Polymerization of Urea for Efficient Sunlight-Driven Photocatalytic Hydrogen Production. *Nanoscale* **2012**, *4*, 5300-5303.

(42) Thomas, A.; Fischer, A.; Goettmann, F.; Antonietti, M.; Müller, J.-O.; Schlögl, R.; Carlsson, J. M. Graphitic Carbon Nitride Materials: Variation of Structure and Morphology and Their Use as Metal-Free Catalysts. *J. Mater. Chem*. **2008**, *18*, 4893-4908.

(43) Dong, F.; Zhao, Z.; Xiong, T.; Ni, Z.; Zhang, W.; Sun, Y.; Ho, W.-K. In Situ Construction of g-C_3N_4/g-C_3N_4 Metal-Free Heterojunction for Enhanced Visible-Light Photocatalysis. *ACS Appl. Mater. Interfaces* **2013**, *5*, 11392-11401.

(44) Vanýsek, P. *Electrochemical Series*; CRC PRESS: USA, 2000.

(45) Hinnemann, B.; Moses, P. G.; Bonde, J.; Jørgensen, K. P.; Nielsen, J. H.; Horch, S.; Chorkendorff, I.; Nørskov, J. K. Biomimetic Hydrogen Evolution: MoS_2 Nanoparticles as Catalyst for Hydrogen Evolution. *J. Am. Chem. Soc.* **2005**, *127*, 5308-5309.

(46) Chou, S. S.; Sai, N.; Lu, P.; Coker, E. N.; Liu, S.; Artyushkova, K.; Luk, T. S.; Kaehr, B.; Brinker, C. J. Understanding Catalysis in a Multiphasic Two-Dimensional Transition Metal Dichalcogenide. *Nat. Commun.* **2015**, *6*, 8311-8318.

(47) Deng, J.; Li, H.; Xiao, J.; Tu, Y.; Deng, D.; Yang, H.; Tian, H.; Li, J.; Ren, P.; Bao, X. Triggering the Electrocatalytic Hydrogen Evolution Activity of the Inert Two-Dimensional MoS_2 Surface Via Single-Atom Metal Doping. *Energy Environ. Sci.* **2015**, *8*, 1594-1601.

(48) Wang, X.; Chen, X.; Thomas, A.; Fu, X.; Antonietti, M. Metal-Containing Carbon Nitride Compounds: A New Functional Organic-Metal Hybrid Material. *Adv. Mater.* **2009**, *21*, 1609-1612.

(49) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **1996**, *77*, 3865-3868.

(50) Grimme, S. Semiempirical GGA-Type Density Functional Constructed with a Long-Range Dispersion Correction. *J. Comput. Chem*. **2006**, *27*, 1787-1799.
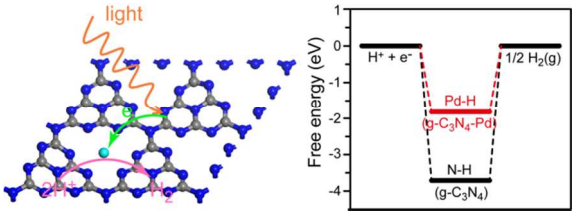
Table of Contents Graphic